PheroSAM-3D Implementation Plan

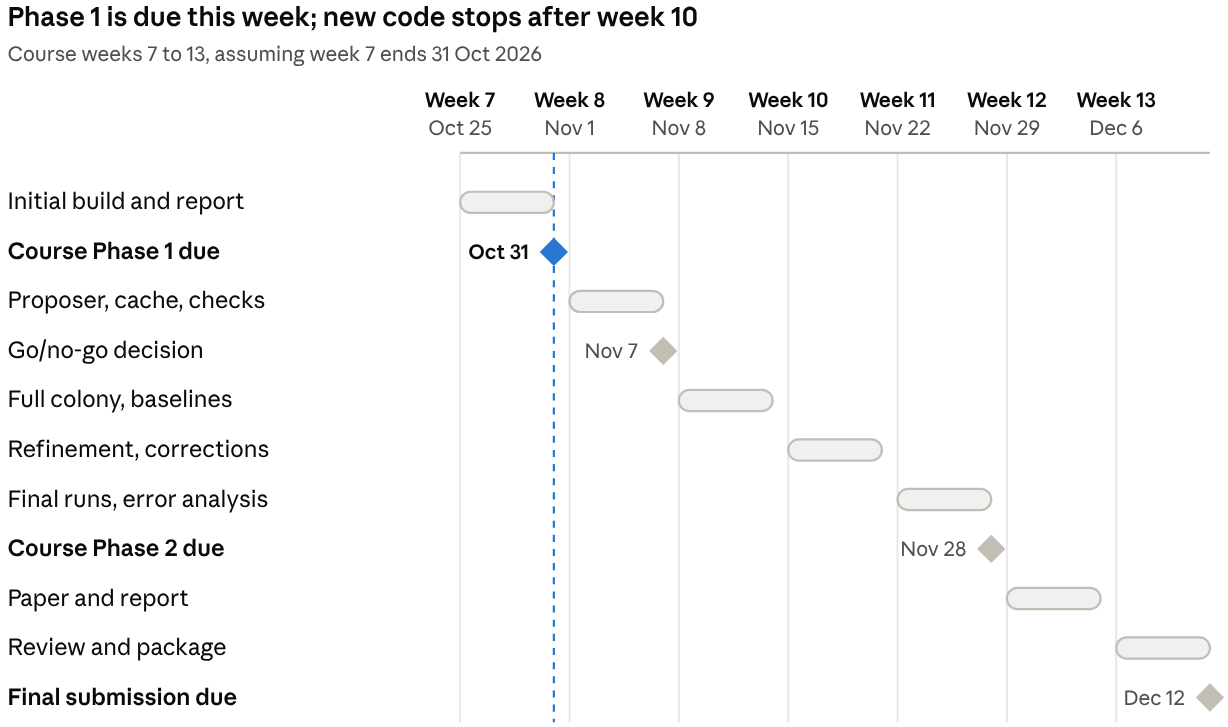
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Overview

You have about 6 weeks, not 14, so the paper becomes a single-organ study on multi-site prostate MRI. The core contribution stays the same: a hybrid ant colony (MMAS + DP seed + ACO_R) picking and refining MedSAM boxes, plus pheromone flags and simulated doctor corrections. Competing colonies for several organs become a stretch goal.

Dates below assume week 7 ends on 31 October 2026 and each later week is 7 days. Replace them with your course's real dates.

Course phase	Week	Assumed due	What you must show
Phase 1: proposal + initial implementation	7	this week	Dataset works, baseline works, NIC designed and partly running
Phase 2: optimization + evaluation	11	about 28 Nov	Complete pipeline producing measurable results, compared with baselines
Phase 3: final code, report/presentation, paper (.docx + .pdf)	13	about 12 Dec	Everything, written up



course timeline · weeks 7–13, 3 deadlines

Weeks 8–10 build the method; week 11 only runs and analyses it; weeks 12–13 only write and package.

In scope:

- Multi-site prostate MRI: proposer trained on one site, tested on the others
- MMAS with DP (Viterbi) seed and ACO_R box refinement, on a frozen MedSAM
- Baselines under equal budget: best box per slice, greedy propagation, DP, random search, GA, PSO, simulated annealing; MedSAM2 if time allows; oracle boxes as upper bound
- Pheromone-entropy flags and simulated doctor corrections
- Error analysis per site and per slice position (top, middle, bottom of the prostate)

Stretch (only if week 10 finishes on time): competing colonies on 3 AMOS CT organs (liver, spleen, right kidney), about 20 volumes.

Future work only: pheromone memory across hospitals, MedSAM2 plug-in mode, cardiac and tumor datasets, the full 15-organ AMOS.

Roles for two people: one owns data and models (preprocessing, nnU-Net proposer, MedSAM cache, metrics, figures); the other owns optimization (fitness, ACO, baseline optimizers, corrections, statistics). Both write.

Week 7 — Course Phase 1 submission (due this week)

Deliverables 1–7 already exist in your proposal; the work this week is deliverables 8–10: a small but real pipeline that loads the data, runs a baseline, and runs a first ant colony. Keep it small: one training site, two test sites, about 10 test volumes.

Day 1 — data and baseline

1. Download the multi-site prostate MRI data. Write a loader that returns a volume, its mask and its site.
2. Preprocess slices exactly as MedSAM's inference code does; run the MedSAM encoder once per slice and cache the embeddings.
3. Proposer, pick one:
   a. **Option A (preferred):** train a small 2D U-Net (for example with MONAI) on the RUNMC site for about an hour. Phase 2 replaces it with nnU-Net.
   b. **Option B (only if no GPU time):** make candidate boxes by randomly shifting and scaling the true box. Label it clearly in the report as a simulated proposer.
4. Baseline: MedSAM with the most confident box on every slice. Also run greedy propagation (each mask gives the next slice's box). Compute 3D Dice per volume.

Day 2 — first NIC results

1. Build 3–5 candidate boxes per slice plus a no-object option; decode each with MedSAM; cache masks and confidence.
2. Fitness with the two easy terms only: confidence + slice-to-slice smoothness.
3. DP (Viterbi) on that fitness.
4. Basic MMAS: discrete choice only (no ACO_R, no shape term yet); 20 ants, 30 rounds.
5. One results table on the 10 test volumes: best box, greedy, DP, MMAS, oracle boxes. One figure: the same slices from greedy vs MMAS.

Day 3 — short technical report (about 6–8 pages) + code

1. Sections 1–7: shorten the proposal; fix the broken equations ("arg arg" should be arg max) and the placeholder references.
2. State the updated scope honestly: single organ is the main study; competing colonies are designed but planned as a stretch goal; pheromone memory is future work.
3. Section 8, Initial implementation: what runs today, a pipeline diagram, the code structure.

4. Sections 9–10, Baseline and initial results: the table and figure from Day 2, plus what is still missing and when it will be done.
5. Code: a clean GitHub repo with a README that runs the Day 2 table with one command.

It is fine if MMAS does not beat DP yet with only the local terms; that is expected, since DP is exact for them. Say so in the report and explain that the global terms (shape, penalties) and refinement come in Phase 2.

Done when:

- [ ] Data loads; MedSAM baseline gives Dice on 10 test volumes
- [ ] DP and basic MMAS run on the same volumes
- [ ] Report (deliverables 1–10) and repo submitted

Week 8 — Real proposer, full cache, go/no-go checks

Goal: by the end of week 8 the full prostate data is cached and you know, with numbers, that the idea has room to win.

Steps

1. Leakage check: read MedSAM's training-data list and record whether it includes your prostate sources. Decide: use, exclude overlapping cases, or report the overlap.
2. Train the 2D nnU-Net v2 proposer on RUNMC only (one fold, shorter-epoch trainer if Colab time is tight). Start this on day 1; it runs while you do the rest.
3. Candidates for all test sites: thresholds 0.3 / 0.5 / 0.7, margins 0 and about 5 px, duplicates removed, $K \leq 5$, plus no-object. Keep each candidate's type label.
4. MedSAM cache for all test volumes: embeddings (float16), candidate masks, predicted IoU, and a stability score as a backup confidence. Time one decoder call; it sets the search budget.
5. Shape term: train a small 3D autoencoder on RUNMC masks (fixed crop, for example 64^3). If it is not working by Thursday, use simple priors instead: volume range, compactness, slice-area profile.
6. Go/no-go checks on the RUNMC validation split:

Check	Pass if	If it fails
Headroom: oracle selection vs best-confidence box, 3D Dice	Gap of about 3 Dice points or more	Make refinement and corrections the main story
Candidate recall: slices with a candidate of IoU ⩾ 0.7	About 80% or more	Report it; refinement must close the gap
Confidence: Spearman of predicted IoU vs true slice Dice	About 0.4 or more	Use the stability score instead
Fitness: Spearman of F vs true 3D Dice over about 200 random paths per volume	Median about 0.5 or more	Reweight or simplify F before week 9

Done when:

- [ ] Leakage decision written down
- [ ] Candidate and mask cache complete for all test sites
- [ ] Go/no-go table filled in and a decision made

Week 9 — Full MMAS with DP seed, and all baselines

Goal: every method runs through one interface, under one budget, on all prostate test sites.

Steps

1. One interface for every optimizer: `optimize(cache, fitness, budget, seed) -> path`. Budget = 600 fitness evaluations per volume (20 ants × 30 rounds) for every method.
2. Full fitness: Conf + Smooth + Shape + Agree − Pen, with the weights chosen in week 8.
3. MMAS, complete:
   a. Pheromone on edges between neighbouring slices, including the no-object node
   b. Only the best ant deposits; iteration-best early, global-best later
   c. τ_min and τ_max from the standard MMAS formulas; clip every round
   d. Stop after 10 rounds without improvement
4. DP seed: Viterbi on Conf + Smooth; its edges start at τ_max, all others at 0.5 × τ_max.

5. Baselines: best box per slice, greedy propagation, DP, random search, GA (index + offset per slice), PSO, simulated annealing. Give GA, PSO and SA the same small tuning effort as ACO.
6. Tune ACO and the baselines on the RUNMC validation split only (for example 20–30 Optuna trials each); freeze the settings in the config.
7. First full run on all test sites with 1 seed; plot fitness vs evaluations for every method.

Done when:

- [ ] All methods run under the same budget
- [ ] First full results table on all test sites
- [ ] Convergence plot drafted

Week 10 — Refinement, ablations, flags and corrections

Goal: the method is complete; this is the last week for new code. From week 11 on, only bug fixes.

Steps

1. ACO_R refinement:
   a. Offset (dx, dy, dw, dh) as a fraction of box size, limited to about ±15%
   b. One archive of the best 10 offsets per slice; new offsets sampled from Gaussians around them, spread $\xi = 0.85$
   c. Round boxes to a 2-pixel grid and cache masks by box, so repeats cost nothing; decode new boxes in one GPU batch per round
2. Ablations (prostate test sites): Ant System vs MMAS; with vs without DP seed; with vs without ACO_R; with vs without Shape term.
3. "Why not DP" sweep: raise the weight of Shape + Pen step by step and record the gap between DP alone and DP-seeded ACO.
4. Flags: pheromone entropy per slice; flag the top 3. Measure how many of the truly worst slices they catch, vs lowest-confidence and random flags.
5. Corrections: the simulated doctor fixes the worst flagged slice with its true box; that slice becomes a fixed node; the colony re-runs about 10 rounds. Record Dice after 1, 2, 3, 5 corrections. Compare with greedy propagation and DP given the same corrections.
6. Stretch, only if steps 1–5 finished by Wednesday: competing colonies on 3 AMOS organs, about 20 CT volumes, compared with independent colonies + a simple overlap fix.

Done when:

- [ ] Full method (MMAS + DP seed + ACO_R) runs on all test sites
- [ ] Ablation, sweep, flag and correction experiments each produce a CSV
- [ ] Code tagged as `v1.0`

Week 11 — Course Phase 2 submission (about 28 Nov)

Goal: final numbers from frozen code, with graphs, tables and an error analysis, submitted with the updated source code.

Steps

1. Final runs from `v1.0`: every method × every test site × 3 seeds. One CSV row per volume, method and seed.
2. Statistics: Wilcoxon signed-rank per volume, full method vs each other method; Holm correction; mean ± std.
3. Tables:
   a. Main results: 3D Dice, HD95, slice consistency, per method and per test site
   b. Ablations
   c. Flag quality and corrections needed to reach a target Dice
   d. Cost: seconds per volume, MedSAM decoder calls, GPU memory
4. Graphs:
   a. Fitness vs evaluations for every optimizer (convergence)
   b. DP vs DP-seeded ACO gap as the global-term weight grows
   c. Dice vs number of corrections
   d. Scatter of label-free fitness vs true Dice
   e. Qualitative slices: greedy vs DP vs full colony, plus the pheromone-entropy map
5. Error analysis, required by the course:
   a. Dice by test site: which scanners hurt most, and does the proposer's recall explain it?
   b. Dice by slice position: apex, mid-gland, base (the ends of the prostate are usually hardest)
   c. Sort the 10 worst volumes into causes: proposer missed the organ; MedSAM mask leaks; wrong no-object choice at the ends; fitness preferred a wrong path. Show one example of each.

6. Phase 2 report: setup, the tables and graphs above, analysis, error analysis, what remains for Phase 3. Reuse it as the paper's Results section later.

Done when:

- [ ] All tables and graphs rebuilt by script from the CSVs
- [ ] Error analysis with examples
- [ ] Phase 2 report and updated code submitted

Week 12 — Writing the paper and the final report

Goal: a complete paper draft and report by the end of week 12, so week 13 is only review, polish and packaging. No new experiments unless the Phase 2 feedback demands one.

Paper, in the course's structure (write in this order):

1. Proposed Method: slice graph, fitness, MMAS + DP seed, ACO_R, flags and corrections. Take it from the code, not the proposal.
2. Materials and Methods: datasets, splits, leakage handling, MedSAM and the proposer, baselines, equal budget, metrics, statistics.
3. Experimental Results: from the Phase 2 report, tightened.
4. Discussion: when the colony helps, when it does not, what the error analysis showed; limitations (one organ, simulated doctor, possible MedSAM data overlap).
5. Related Work: MedSAM and SAM in 3D, automatic prompting (including the 2026 GA + SAM paper), ACO and matheuristics.
6. Introduction, then Conclusion, then Abstract and Keywords last.
7. References: check every one against the real paper; the proposal has placeholders and one "from memory" entry.

Final report / presentation: it follows the course's own list (Introduction through Future work), and mostly reuses paper text. Build the slides from the paper's figures: about 12–15 slides, one figure per result.

Share the draft with your instructor or a classmate by Thursday so you have feedback for week 13.

Done when:

- [ ] Full paper draft, all sections
- [ ] Report or slide deck drafted
- [ ] Draft sent for feedback

Week 13 — Final submission (about 12 Dec)

Goal: three packages handed in, each checked by the person who did not write it.

A. Source code. The repo must have one folder or script per course item, and a README with the command for each:

Course item	What it is in this project
Data preprocessing	Loading, reorienting, normalizing and slicing the prostate volumes
Baseline model	MedSAM with best box per slice, and greedy propagation
NIC algorithm	MMAS + DP seed + ACO_R; GA, PSO, SA for comparison
NIC-AI integration	Candidate boxes → MedSAM decoder → fitness → ants
Training	nnU-Net proposer and the shape autoencoder
Testing	Running every method on the test sites
Evaluation	Dice, HD95, consistency, statistics
Visualization	Every figure and table, rebuilt by script

B. Final report / presentation, in the course's order: Introduction, Problem statement, Related work, Methodology, Proposed NIC approach, AI architecture, Experimental setup, Results, Discussion, Limitations, Conclusion, Future work.

C. Research paper as both .docx and .pdf. Write it in Word or export to Word at the end; produce the PDF from that same file so the two match. Check equations, figure numbers and references after export.

Final checks:

- [ ] Fresh clone of the repo runs the main results table
- [ ] Every number in the paper matches a CSV
- [ ] Report or slides follow the course's section list
- [ ] Paper submitted as .docx and .pdf
- [ ] Decide whether to submit the paper to an outside venue after the course

Deliverables map

Every course item for Phases 1 and 2 has a source in this plan; Phase 3′s items are mapped in the week 13 section.

Course phase	Item	Comes from
1	Title, problem definition, dataset, literature review	Proposal sections 1–5, shortened and fixed
1	Selected NIC algorithm, AI model, architecture	Proposal sections 6–8, with the updated scope
1	Initial implementation	Week 7, days 1–2
1	Baseline model	Week 7: MedSAM best box per slice + greedy propagation
1	Initial experimental results	Week 7: table on 10 test volumes + one figure
2	Complete NIC implementation	Weeks 9–10: MMAS + DP seed + ACO_R
2	Complete AI model	Week 8: nnU-Net proposer, MedSAM cache, shape autoencoder
2	NIC-AI integration	Weeks 8–10: candidates → MedSAM → fitness → ants
2	Optimization experiments	Week 10: ablations, DP-vs-ACO sweep, tuning
2	Baseline vs NIC comparison	Weeks 9 and 11: equal-budget comparison, 3 seeds, statistics
2	Evaluation metrics	Dice, HD95, slice consistency, overlap, corrections to target, runtime
2	Graphs, tables, analysis	Week 11, steps 3–4
2	Error analysis	Week 11, step 5
2	Updated source code	Tagged `v1.0` at the end of week 10

Risks and fallbacks

With 6 weeks there is no slack for a second attempt, so each risk has a cut that keeps the paper intact.

Risk	Early warning	Fallback
Phase 1 time runs out	Day 2 not finished	Submit with DP + baseline only and MMAS "designed, partly implemented"; the course allows this
nnU-Net training too slow	Not done by mid-week 8	Keep the week 7 small U-Net as the proposer
Fitness does not track true Dice	Week 8 Spearman below about 0.5	Drop the shape autoencoder, use simple priors; reweight terms
Small headroom	Week 8 gap under about 3 Dice points	Make refinement and corrections the main result
Simulated annealing ties ACO	Week 9 results	Argue from the flags and correction propagation, which SA does not give
GPU quota runs out	Week 8 timing × planned runs	Fewer ants and rounds for every method equally; fewer test volumes
Week 10 overruns	Corrections not done by Thursday	Drop the stretch goal; cut corrections to 1, 2, 3
Exams or other courses collide	Check the calendar now	Move writing into week 11 evenings; run long jobs overnight

Scope rule: any new idea goes into Future work, not into the code.